

Eindhoven University of Technology

AI-Supported Sustainability Self-Assessment for TU/e Research

ENSU – Written Assignment Stage 1

Course: 5ARF0 Ethics of AI: Theory and Practice
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Main-text count excludes headings, tables, figures, references, appendices and draft placeholders.

1 Introduction and Motivation

1.1 Topic, Motivation, Values and Needs

[TODO: Define the sustainability problem in TU/e research. Explain why an AI-supported tool may help. Cite evidence.]

Values and needs. [TODO: Introduce stakeholder needs and the main values: sustainability, transparency, accountability, fairness, and human oversight.]

1.2 Objectives and Key Terms

[TODO: State one project aim and one research question. Define the essential terms.]

1.3 Context and Scope

[TODO: Specify users, research activity, decision point, setting, and non-goals.]

1.4 Tools, Methods and Frameworks

[TODO: Explain why Design for Values or Value Sensitive Design fits the aim. Introduce conceptual, empirical, and technical investigations with references.]

Logic: research context → stakeholder need → value → requirement → tool feature → success indicator.

2 Initial Results

2.1 Preliminary Literature Search

[TODO: Report databases, search terms, dates, selection criteria, number of sources, and main evidence gaps.]

Table 1: Literature-search log.

Source	Type	Main finding	Need or value	Design implication
[TODO: Citation]	[TODO: Type]	[TODO: Finding]	[TODO: Need/value]	[TODO: Implication]
[TODO: Citation]	[TODO: Type]	[TODO: Finding]	[TODO: Need/value]	[TODO: Implication]

Preliminary findings. [TODO: Synthesize researcher needs, current assessment methods, AI opportunities, known risks, and evidence gaps.]

2.2 Design Strategy

[TODO: Describe the steps from evidence and stakeholder input to requirements, mock-up, expert review, evaluation, and revision.]

2.3 Initial Requirements and Constraints

[TODO: Separate functional, non-functional, ethical requirements, constraints, and non-goals.]

Table 2: Requirement traceability matrix.

ID	Stakeholder need	Value	Requirement	Tool feature	Success indicator
R1	[TODO: Need]	Sustainability	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]
R2	[TODO: Need]	Transparency	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]
R3	[TODO: Need]	Human over-sight	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]
R4	[TODO: Need]	[TODO: Value]	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]

2.4 Anticipated Function and Purpose of the Tool

[TODO: Describe the user task, inputs, AI function, outputs, next human decision, and connection to the current TU/e self-assessment tool. State how experts can modify it.]

Figures 1 and 2 present a design proposal for group confirmation, rather than implemented functionality. AI would structure descriptions and draft source-supported suggestions; a calculation module would apply explicit formulas. Missing inputs would be returned for clarification, with unsupported estimates withheld if the information remains unavailable.

A proposed TU/e sustainability case archive would retain activity conditions, the model and recommendation versions used, and follow-up evidence under a pseudonymous case identifier. For a teaching-material printing case, it could record print volume and settings, the advice adopted, and subsequent paper use alongside any assessment-score change. Comparable cases could then inform peer comparisons and revised suggestions. A higher score alone would not establish environmental improvement. Researchers would confirm outcomes and consent to reuse; a campus curator would review evidence and comparability before admitting cases or revising guidance. This extension to teaching activities would require client agreement on scope.

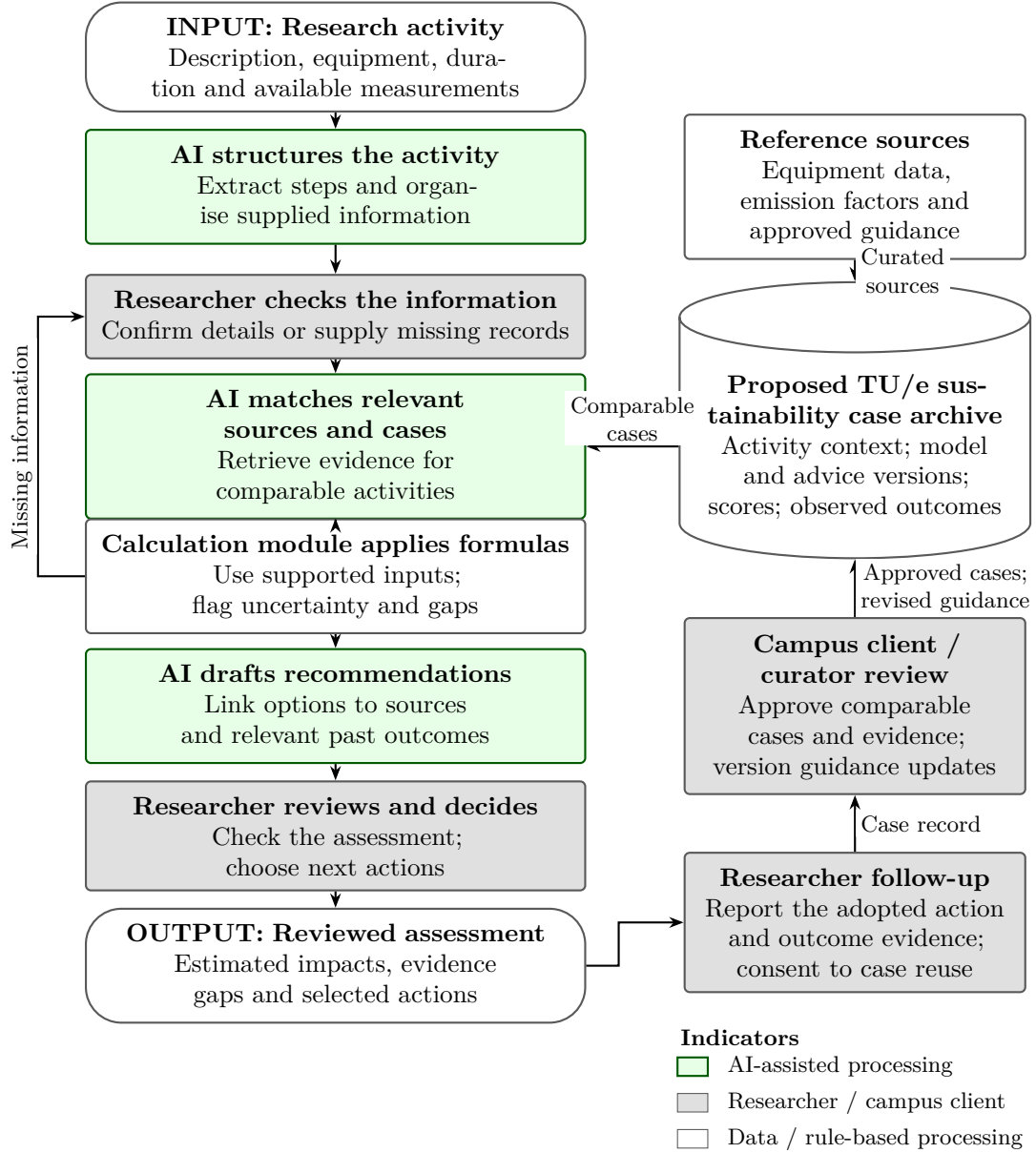


Figure 1: Proposed assessment workflow with a campus case archive. Reviewed follow-up evidence informs later case comparisons and guidance updates. The archive and workflow are design proposals, not implemented services.

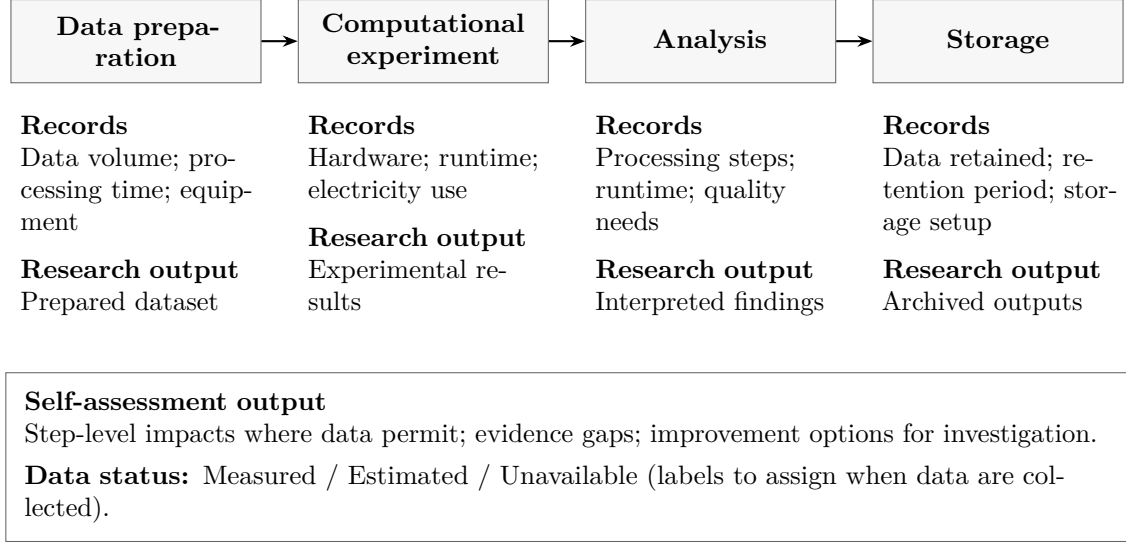


Figure 2: Illustrative computing activity and its assessment records. Research outputs are shown separately from self-assessment outputs; no measurements are assumed.

Illustrative assessment boundary: selected computing activities. Hardware manufacturing and shared infrastructure require additional data.

2.5 Projected Users, Stakeholders and Values

[TODO: Identify users, affected groups, governors, experts, and their roles in the design.]

Table 3: Stakeholder and value map.

Stakeholder	Need	Value	Benefit or harm	Design role
TU/e researchers	[TODO: Need]	[TODO: Value]	[TODO: fit/harm]	[TODO: Role]
[TODO: Stakeholder]	[TODO: Need]	[TODO: Value]	[TODO: fit/harm]	[TODO: Role]
[TODO: Stakeholder]	[TODO: Need]	[TODO: Value]	[TODO: fit/harm]	[TODO: Role]

3 Discussion

3.1 Ethical Issues and Value Conflicts

The definition of sustainable research determines how activities are assessed. Hasselbalch and van Wynsberghe [2025] locate sustainability within sociotechnical relations of power, making the selection of criteria an ethical issue. If electricity consumption is prioritised, an intensive experiment may be rated unfavourably despite its scientific value. In the proposed assessment, environmental indicators would therefore be presented alongside explanations of research purpose. Comparisons would be restricted to activities with comparable objectives and quality requirements. Client agreement is needed on the weighting of criteria and the appropriateness of comparisons across disciplines. A single aggregate score could otherwise conceal disagreements about research priorities.

Numerical estimates may be treated as more authoritative than their evidence warrants. Lacoste et al. [2019] identify hardware, runtime and electricity supply as factors affecting emissions from machine learning training. Their method does not establish validity for every research activity. Apparent differences between cases may be smaller than calculation uncertainty. Measured and estimated values would therefore be distinguished, with missing information flagged and sources and boundaries disclosed. Input correction would remain available to researchers. These provisions support scrutiny, but cannot ensure that an assessment is critically examined.

The resource consumption of the assessment tool also requires justification. Lin [2023] argues for transparency about AI’s energy use. Explicit calculations would be assigned to fixed formulas, while AI assistance would be limited to interpreting descriptions and drafting supported suggestions. Comparison with a structured non-AI assessment is needed to establish whether any additional benefit warrants the computation and review effort. Environmental improvement has not yet been demonstrated.

3.2 Design Barriers and Challenges

Defining a manageable assessment boundary is a practical constraint. Consumption from shared equipment and computing services may be difficult to attribute to individual activities. One representative workflow would be mapped with the client to establish which records are available. Estimates of equipment operation would state their exclusions. Replacement advice would require evidence about manufacturing impacts as well as operating savings.

Reference data and recommendations require maintenance. Sources, review dates and version histories would be recorded so that earlier assessments remain traceable. Following the institutional perspective of Hasselbalch and van Wynsberghe [2025], responsibility for correcting or withdrawing outdated guidance would need to be assigned at TU/e. The responsible role and available resources remain unresolved. For the proposed campus archive, outcome evidence and comparable activity conditions would require review before cases were reused. Higher assessment scores alone would not demonstrate reduced environmental impact. Reported outcomes would be distinguished from verified measurements, since selective reporting could favour successful cases and bias recommendations for future users of the proposed archive.

Evaluation would compare AI-assisted feedback with a structured checklist on the same cases. Evidence support, omitted issues and correction effort would be examined to establish whether further development is justified.

Table 4: Anticipated ethical risks and proposed responses. References motivate the risks; responses are project-specific proposals.

Risk	Stakeholders	Values	Response and basis	Limitation
Privacy	Researchers; case contributors	Privacy; autonomy	Minimise stored details; control archive access and case reuse [Hasselbalch and van Wynsberghe, 2025].	Activity descriptions may identify a contributor.
Unclear AI advice	Researchers	Transparency; accountability	Show sources, model version and uncertainty [Lacoste et al., 2019].	A retrieved case may be irrelevant or misinterpreted.
Greenwashing	TU/e; researchers	Sustainability; trust	State boundaries; check observed impacts alongside score changes [Lin, 2023].	Better scores need not reflect lower overall impacts.
Rebound effects	TU/e; affected communities	Environmental impact	Track total activity and resource use alongside efficiency [Luccioni et al., 2025].	Savings may encourage additional activity.
Unfair recommendations	Research groups	Fairness	Compare similar objectives and conditions; review archive coverage [Hasselbalch and van Wynsberghe, 2025].	Uneven case contributions can distort comparisons.
Reduced human control	Researchers; campus curator	Human oversight	Keep decisions with researchers; assign responsibility for archive updates [Santoni de Sio and van den Hoven, 2018].	Formal review does not ensure critical engagement.

References

- Antoine Berthelot, Eddy Caron, Maxime Jay, and Laurent Lefèvre. Understanding the environmental impact of generative ai services. *Communications of the ACM*, 68(7):46–53, 2025. doi: 10.1145/3725984.
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- Patrick K. Lin. The cost of training a machine: Lighting the way for a climate-aware policy framework that addresses artificial intelligence’s carbon footprint problem. *Fordham Environmental Law Review*, 34(2), 2023. URL <https://ir.lawnet.fordham.edu/elr/vol34/iss2/1/>.
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- Filippo Santoni de Sio and Jeroen van den Hoven. Meaningful human control over autonomous systems: A philosophical account. *Frontiers in Robotics and AI*, 5:15, 2018. doi: 10.3389/frobt.2018.00015.

A Documentation of Process

[TODO: Record meetings, research, design work, writing work, decisions, and evidence.]

Table 5: Project process log.

Date	Participants	Activity	Output or decision	Next action
18 Sep 2026	Yehao Chen; AI writing assistance	Revise discussion; diagrams	Dis-draft Proposed workflow; revised text	Group review of scope and wording
[TODO: Date]	[TODO: Names]	[TODO: Activity]	[TODO: Output]	[TODO: Action]

B Project Planning

Table 6: Proposed remaining project schedule (2026).

Date	Activity	Owner	Deliverable
18 Sep, 23:59	Integrate and submit Phase 1	[TODO: Owner]	Phase 1 report
22–29 Sep	Confirm scope and evidence needs during coaching	[TODO: Owner]	Revised requirements
30 Sep–10 Oct	Develop mock-up; prepare evaluation and practice pitch	[TODO: Owner]	Mock-up and evaluation plan
11 Oct, 23:59	Submit Phase 2 draft for peer review	[TODO: Owner]	Draft report
16 Oct, 23:59	Complete individual peer review	Each member	Peer feedback
17–20 Oct	Revise and record pitch	[TODO: Owner]	Video pitch: 20 Oct, 23:59
23 Oct, 23:59	Submit Phase 2 final version	[TODO: Owner]	Final report
24 Oct, 23:59	Complete groupmate evaluation	Each member	Individual evaluation